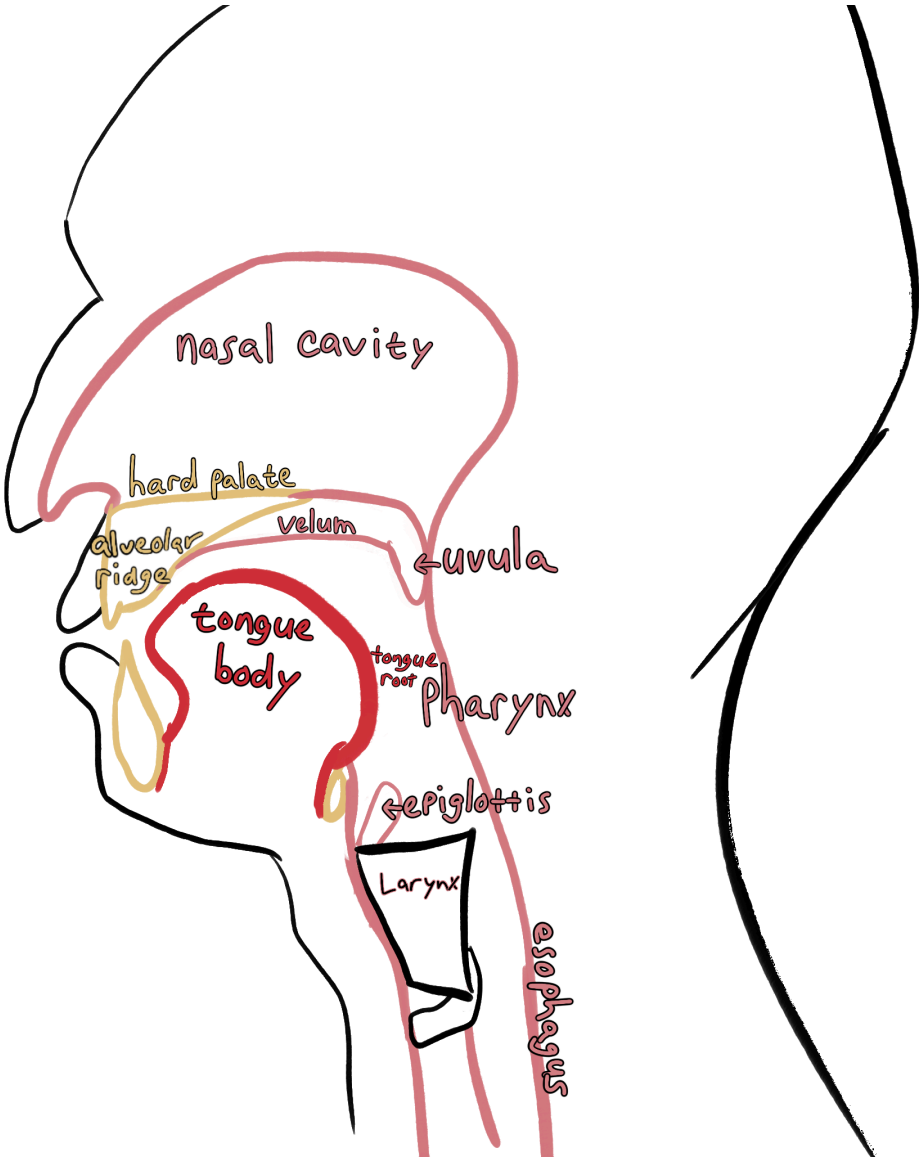


# Speech Anatomy 4

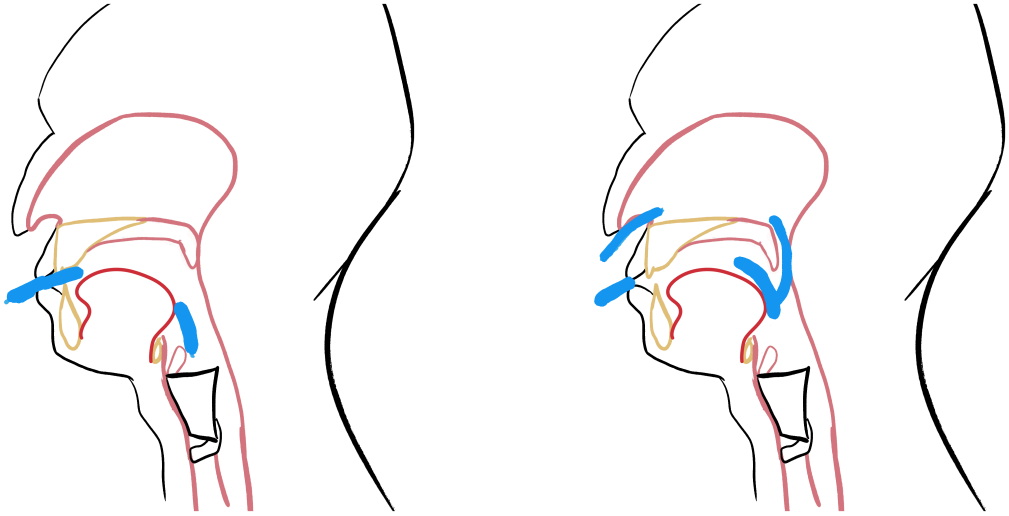
## Supralaryngeal Vocal Tract

Dr. Josef Fruehwald

2026-09-01


nasal cavity
hard palate
alveolar ridge
velum
uvula
tongue body
tongue root
pharynx
epiglottis
Larynx
esophagus

Nasopharynx



## The Tongue



## The Tongue

- **extrinsic** muscles: muscles attached to the tongue
- **intrinsic** muscles: muscles within the tongue

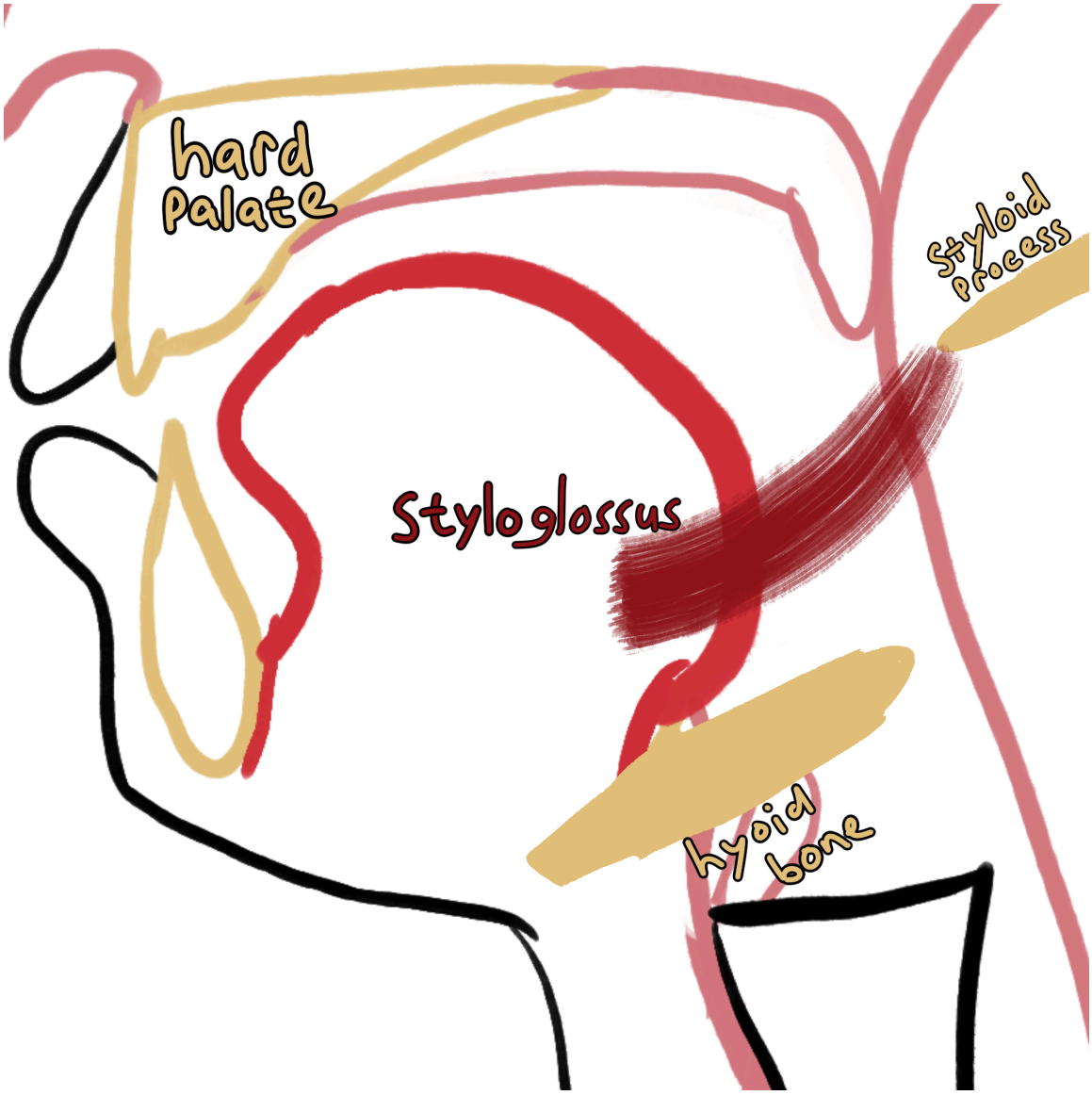
## The Tongue

Muscles are usually named for the what they attach to.

- “-glossus”: Attaching to the tongue

# The Tongue

## Extrinsic Muscles



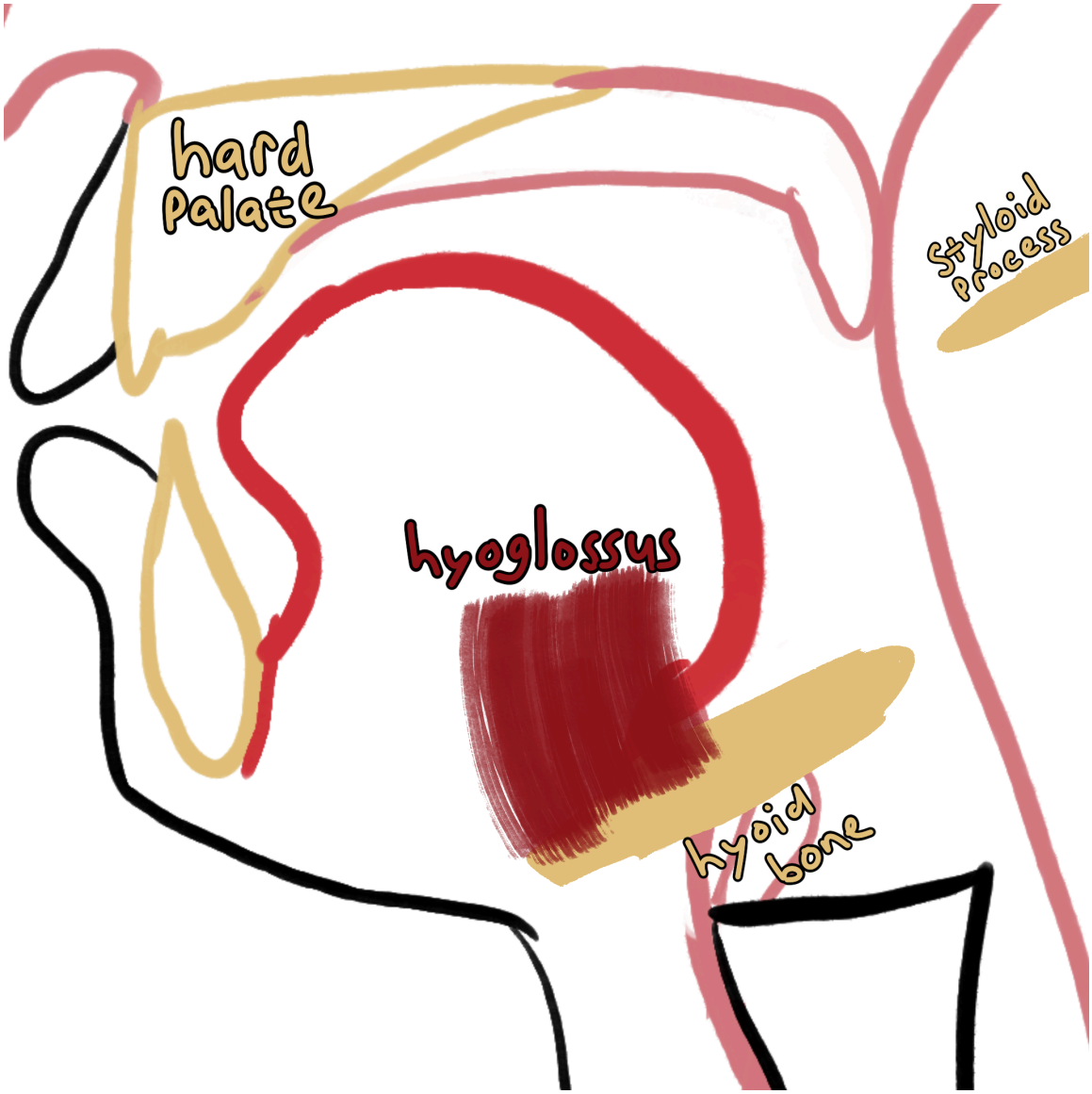

Pulls tongue body back

Styloid Process



# The Tongue

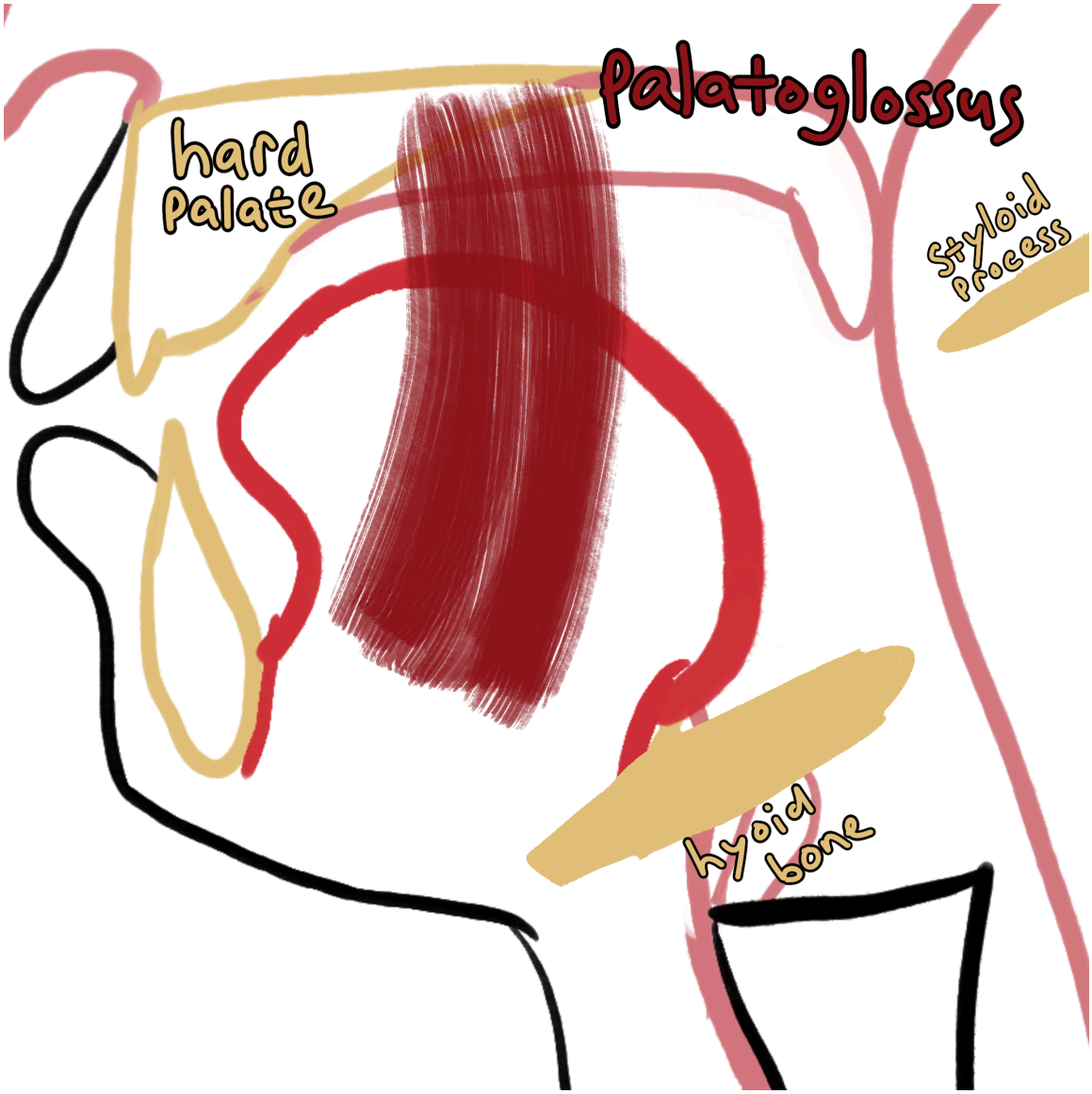
## Extrinsic Muscles




Pulls tongue body down

# The Tongue

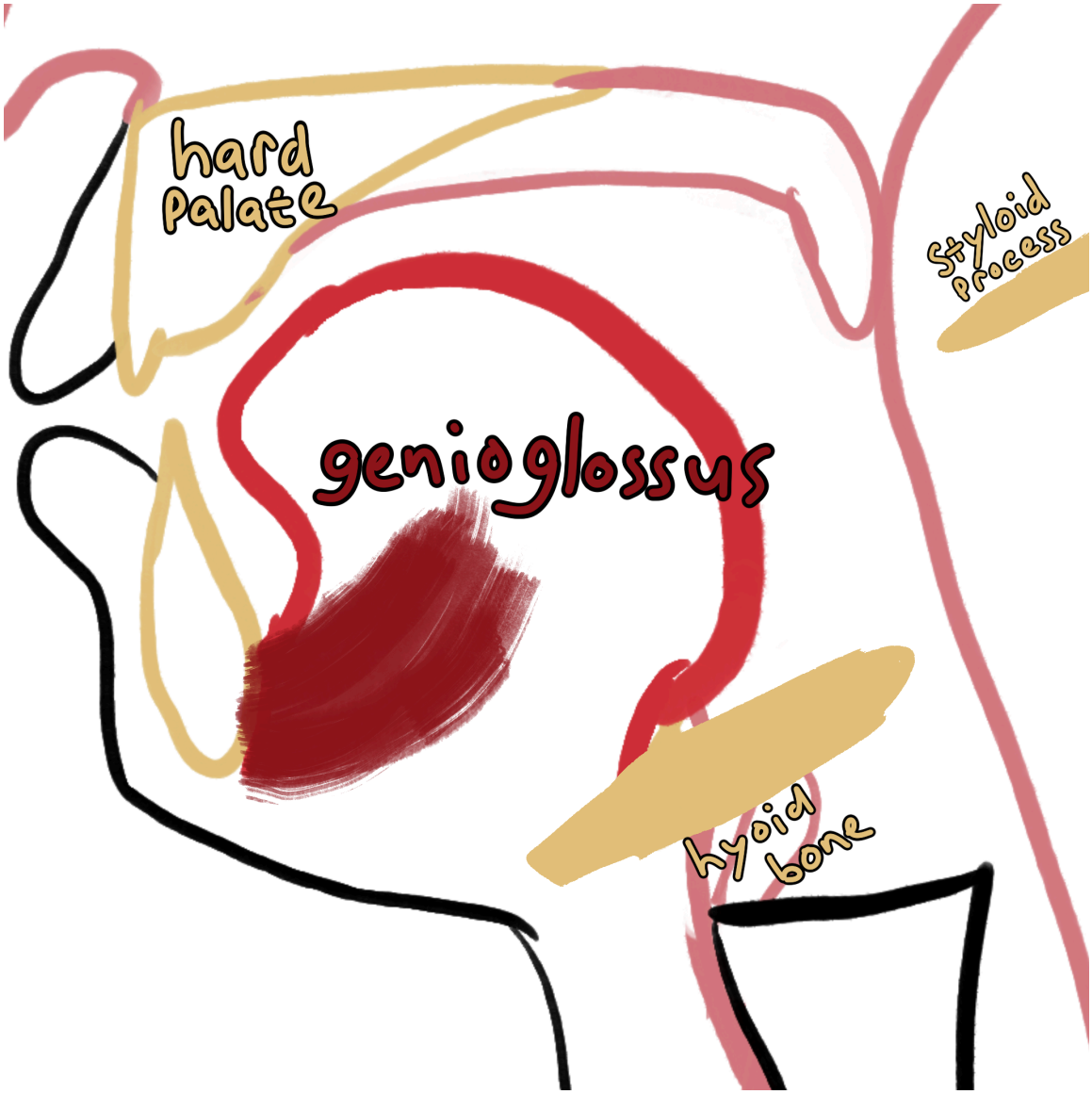
## Extrinsic Muscles




Pulls tongue up

# The Tongue

## Extrinsic Muscles




Downward and forward

## Coordinating it all

Everything is connected to everything

- Hyoglossus pulls down on tongue ?
- Tongue pulls on palatoglossus ?

- Palatoglossus pulls on soft palate

Low vowels are more likely to be nasalized than high vowels.

## Coordinating it all

### Timing

